

Homework 1 Reference Answers

Course: Data Structures

Teacher: Li Qiang

File id: mock-wjid-102 (mock=102)

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Exercise 1 (linear list)

A singly linked list with a head node supports insertion at the front in $O(1)$ and deletion by value in $O(n)$. The reference answer keeps a tail pointer so that appending is also $O(1)$.

Exercise 3 (complexity)

The loop executes $n + (n-1) + \dots + 1$ times, therefore $T(n) = O(n^2)$.

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Exercise 5 (implementation notes)

Remember to release every node that is unlinked; the reference implementation returns the removed element and leaves the list valid.

Grading notes

Full credit requires the complexity argument, not only the code.